

1. What is a Python library?

Python libraries are collections of pre-written code that help perform tasks easily without writing everything from scratch.

Example: Instead of writing code for calculations, you can use the **NumPy** library.

2. Why Libraries are Important?

- Save time
- Reduce code complexity
- Improve performance
- Used in real-world AI/ML projects

3. What is math module?

The math module is a built-in Python library that provides mathematical functions to perform calculations easily.

It helps you work with:

- Basic math operations
- Trigonometry
- Logarithms
- Constants like π and e

Key Features

- No installation required (comes with Python)
- Faster and more accurate calculations
- Useful in scientific and data-related tasks

4. What is random module used for?

The random module in Python is used to generate random values for different purposes.

Common Functions

- `randint()` → random integer
- `random()` → random float (0 to 1)
- `choice()` → pick random item
- `shuffle()` → mix list

5. What is data visualization?

Data visualization is the process of representing data using charts, graphs, and visuals so that information becomes easier to understand.

Why is Data Visualization Important?

- Makes complex data easy to understand
- Helps identify patterns and trends
- Supports better decision-making
- Used in business, science, and AI/ML